Compiler Construction Lab

Gull e Zahra (54154)

Lab 7

Task 1

```
#include <iostream>
#include <string>
using namespace std;

string s;
int pos = 0;

void S() {
   if (pos < s.length() && s[pos] == 'a') {
      pos++;
      S();
      if (pos < s.length() && s[pos] == 'b') {
         pos++;
      } else {
         cout << "Rejected";
         exit(0);
      }
   }
}

int main() {
   cin >> s;
   S();
   if (pos == s.length())
      cout << "Accepted";
   else
      cout << "Rejected";
}
```

C:\Users\malik\Documents\t1l7.exe

```
aaabbb
Accepted
--------------------------------
Process exited after 60.04 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\malik\Documents\t1l7.exe
aaab
Rejected
--------------------------------
Process exited after 5.428 seconds with return value 0
Press any key to continue . . .
```

Recursion Explanation

Input String: aaabbb

- Step 1: Start with S. The first letter is a, so use the rule: $S \rightarrow aSb$.
- Step 2: Look at the remaining middle part. The input still has another a, so again use: $S \rightarrow aSb$.
- Step 3: The input has one more a, so use the rule a third time: $S \rightarrow aSb$.
- Step 4: Now after using three as, no more a is left to process in the middle. It becomes: $a(a(aSb)b)b$. No a is left, so: $S \rightarrow \varepsilon$.
- Step 5: $a(a(a\varepsilon b)b)b$. Since ε means nothing, it becomes: aaabbb.
- Step 6 (Final String): aaabbb matches the input.

Task 2

```
#include <iostream>
#include <string>
using namespace std;

string s;
int pos = 0;

void E(); void Ep(); void T(); void Tp(); void F();

void E() {
   T();
   Ep();
}

void Ep() {
   if (pos < s.length() && s[pos] == '-') {
      pos++;
      T();
      Ep();
```

```
    }
}

void T() {
    F();
    Tp();
}

void Tp() {
    if (pos < s.length() && s[pos] == '*') {
        pos++;
        F();
        Tp();
    }
}

void F() {
    if (pos < s.length() && s[pos] == 'i') {
        pos++;
    } else if (pos < s.length() && s[pos] == '(') {
        pos++;
        E();
        if (pos < s.length() && s[pos] == ')') {
            pos++;
        } else {
            cout << "Rejected";
            exit(0);
        }
    } else {
        cout << "Rejected";
        exit(0);
    }
}

int main() {
    cin >> s;
    E();
    if (pos == s.length())
        cout << "Accepted";
    else
        cout << "Rejected";
}
```

```
C:\Users\malik\Documents\t1l7.exe
i-i*i
Accepted
--------------------------------
Process exited after 21.11 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\malik\Documents\t1l7.exe
i-*i
Rejected
--------------------------------
Process exited after 9.997 seconds with return value 0
Press any key to continue . . .
```

TASK 3

```
#include <iostream>
#include <string>
using namespace std;

string s;
int pos = 0;

void E(); void Ep(); void T(); void Tp(); void F();

void E() {
   T();
   Ep();
}

void Ep() {
   if (pos < s.length() && (s[pos] == '+' || s[pos] == '-')) {
      pos++;
      T();
      Ep();
   }
}

void T() {
   F();
```

```
    Tp();
}

void Tp() {
    if (pos < s.length() && (s[pos] == '*' || s[pos] == '/')) {
        pos++;
        F();
        Tp();
    }
}

void F() {
    if (pos < s.length() && s[pos] == 'i') {
        pos++;
    } else if (pos < s.length() && s[pos] == '(') {
        pos++;
        E();
        if (pos < s.length() && s[pos] == ')') {
            pos++;
        } else {
            cout << "Rejected";
            exit(0);
        }
    } else {
        cout << "Rejected";
        exit(0);
    }
}

int main() {
    cin >> s;
    E();
    if (pos == s.length())
        cout << "Accepted";
    else
        cout << "Rejected";
}
```

```
C:\Users\malik\Documents\t1l7.exe
(i+i)*i/i
Accepted
--------------------------------
Process exited after 83.74 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\malik\Documents\t1l7.exe
i++i
Rejected
--------------------------------
Process exited after 6.346 seconds with return value 0
Press any key to continue . . .
```

Parsing Steps

Input String: (i + i) * i / i

- Step 1: Start from E. $E \rightarrow TE'$. Go to T, then F.
- Step 2: F sees (, so it matches (and starts a new E for the inside.
- Step 3: Inside brackets, match i, then E' matches +, then match the second i.
- Step 4: F matches the closing).
- Step 5: Now back in the outer T', match *, then F matches the third i.
- Step 6: T' still has input, so match /, then F matches the fourth i.
- Step 7: All input is consumed.